LIPID PEROXIDATION

INTRODUCTION

- Oxidative deterioration of lipids containing any number of carbon-carbon double bonds
- Fatty acids
- Cholesterol

- The deleterious effects are considered to be caused by **FREE RADICALS** produced during peroxide formation from fatty acids containing double bonds, ie, those found in the naturally occurring polyunsaturated fatty acids

Free radicals

A **free radical** is any species capable of independent existence that contains one or more **UNPAIRED ELECTRONS**

Free radicals easily react with a biological molecules (mainly non-radicals) generating new radicals – initiate **chain reactions**

FREE RADICAL SOURCES

- Exogenous
  - Foods
  - Air pollutants
  - Radiation
- Endogenous
  - Metabolism (mitochondria and peroxisomes)
  - Detoxification – cytochrome P450
  - Immune cells

PRODUCTION OF TOXIC COMPOUNDS

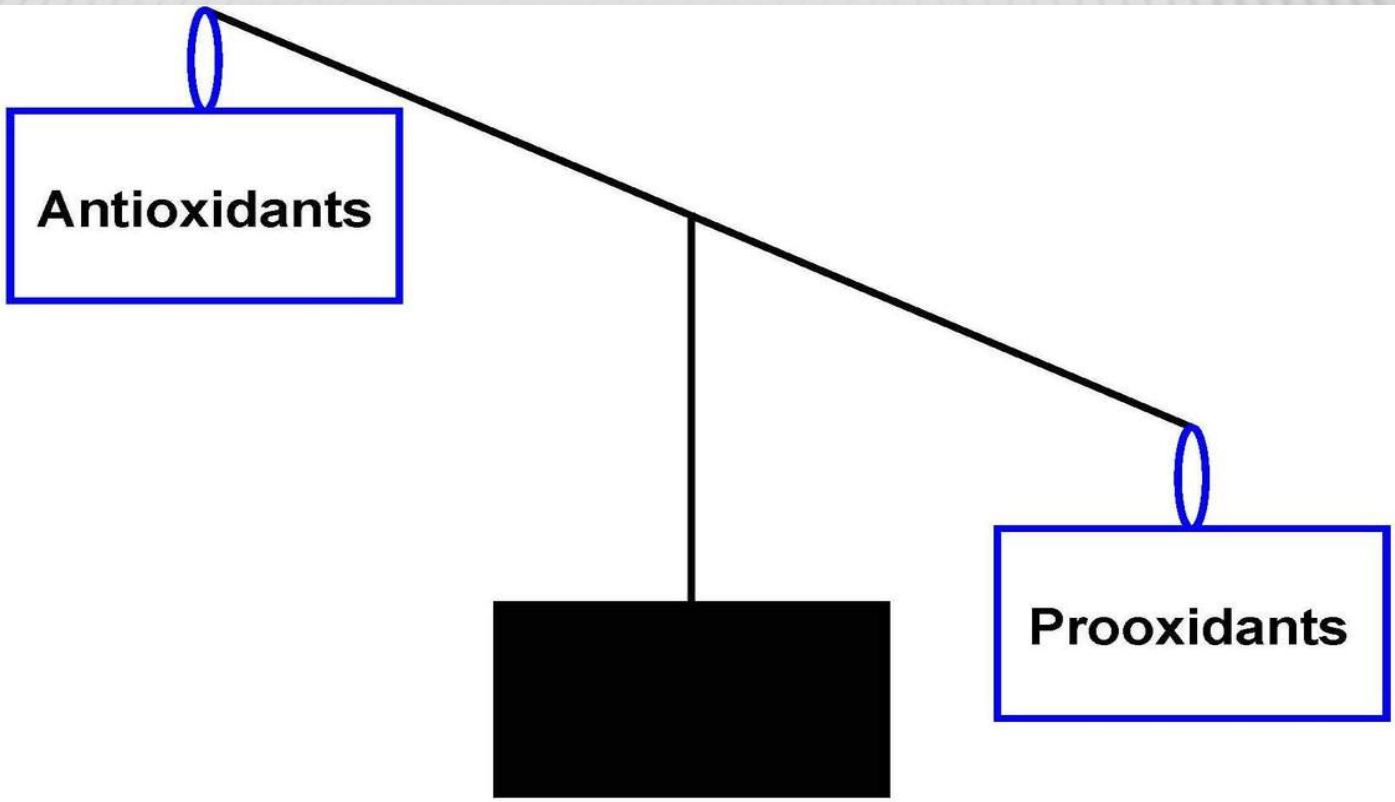
- Many secondary by-products of lipid oxidation are potential carcinogens
- Hydroperoxides are known to damage DNA
- Carbonyl compounds may affect cellular signal transduction
- Aldehydes: 4-hydroxynonenal, and MDA
- Epoxides and hydrogen peroxide by-products are known carcinogens

REACTIVE OXYGEN SPECIES AND FREE RADICALS

- **Reactive Oxygen Species**
  - Triplet oxygen
  - Superoxide
  - Singlet Oxygen
  - Hydroperoxyl radical
  - Hydroxyl radical
  - Hydrogen peroxide
  - Ozone
- **Peroxyl radical (ROO.)**
- **Alkoxyl radical (RO.)**
- **Iron-oxygen complexes**
- **Thiyl radicals (RS.)**
- **Nitric oxide (.NO)**

IMPLICATIONS OF LIPID OXIDATION TO HUMAN HEALTH

OXIDATIVE STRESS



"An imbalance favoring prooxidants and/or disfavoring antioxidants, potentially leading to damage" -H. Sies

PATHOLOGICAL CONDITIONS THAT INVOLVE OXIDATIVE STRESS

- Inflammation
- Atherosclerosis
- Ischemia/reperfusion injury
- Cancer
- Aging

INFLAMMATION



"Vicious cycle" of inflammation and damage:

Immune cells migrate to the site of inflammation

↓

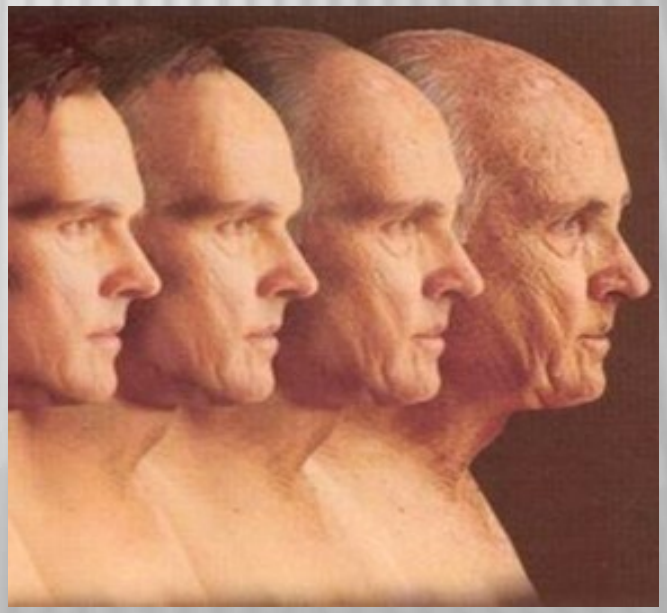
Produce oxidants to kill bacteria

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Oxidants directly damage tissue → (back to: Immune cells migrate to the site of inflammation)

AGING

- The process(es) that occur during life which culminate in changes that decrease an individual's ability to handle biological challenges



FREE RADICALS AND AGING

- Protein cross-linking
  - Wrinkles in skin
  - Reduces flexibility in arterial cells
  - Hampers the functions of proteins and enzymes
  - Impedes message transmission between nerve cells in the brain
- Damage DNA and RNA molecules
  - Results in impaired DNA replication and protein assembly
  - Causes chronic degenerative diseases and cancer
  - Protein degradation

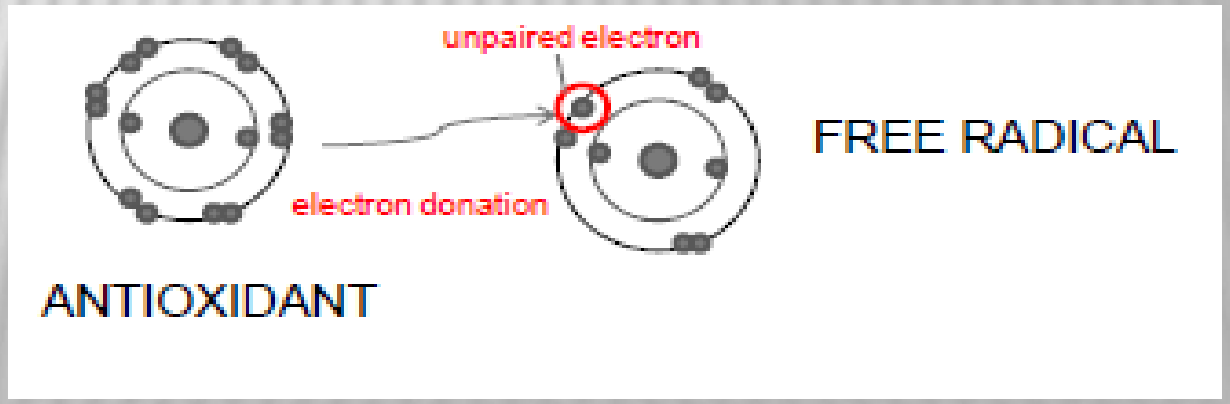
Antioxidative defense

An **antioxidant** is any substance that delays, prevents or removes oxidative damage to a target molecule

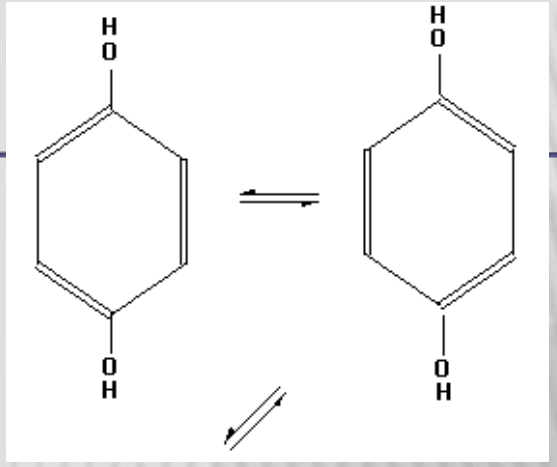
There is *no universal best* antioxidant!

Their relative importance depends upon:

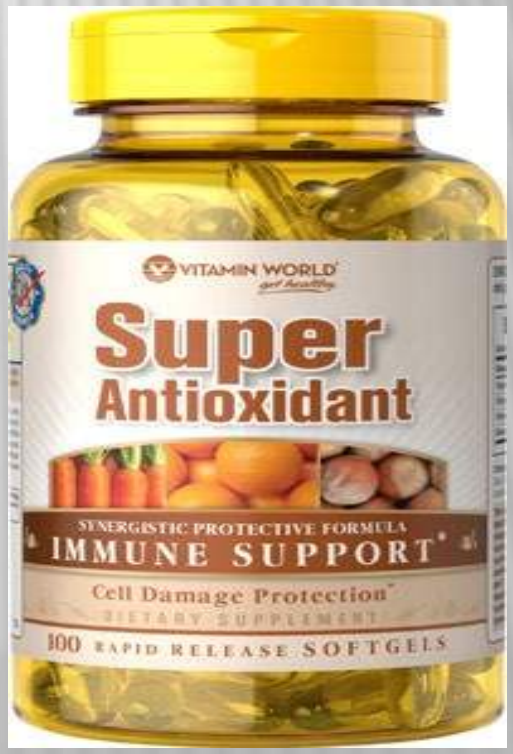
Which, how, where ROS is generated and what target of damage is measured



ANTIOXIDANTS



- Antioxidants function by interfering with the chain reaction.
- If the number of free radicals can be kept low enough, oxidation will not occur.



MAIN MECHANISMS FOR INHIBITION OF OXIDATIVE REACTIONS

1. Interrupt the free-radical chain mechanism

2. Function as being preferentially oxidized - poor protection

3. Reducing agents

4. Chelating agents for free iron

ANTIOXIDANT DEFENSES IN BIOLOGICAL SYSTEMS

- Fat-soluble cellular membrane consists
  - Vitamin E
  - beta-carotene
  - Coenzyme Q (10)

- Water soluble antioxidant scavengers
  - Vitamin C
  - Glutathione peroxidase,
  - Superoxide dismutase
  - Catalase

Thank you